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What is the Candidate Elimination process. Explain. (03 Marks)

The key idea in Candidate-Elimination algorithm is to output a description of the set of all hypotheses consistent with the training examples. The Candidate-Elimination algorithm represents the version space by storing only its most general members (denoted by G) and its most specific members (denoted by S).

(father ?name) (person \$?name) In this why the \$ and ? is used in CLIPS language. (03 Marks)

In CLIPS (C Language Integrated Production System), the \$ and ? characters are used to distinguish between different types of variables used in patterns and templates.

What are the different knowledge analysis process. (Aisa hi kuch tha) (03 Marks)

Knowledge analysis

The goal of knowledge analysis is to analyze and structure the knowledge gained during the knowledge acquisition phase. The key steps to be followed during this stage are

- Identify specific knowledge elements, at the level of concepts, entities, etc.
- From the notes taken during the interview sessions, extract specific
 - o Identify strategies (as a list of points)
 - o Translate strategies to rules
 - o Identify heuristics
 - o Identify concepts

What is aggregation? Is the output is necessary in fuzzy system in aggregation. (03 Marks)

Aggregation is the process by which the fuzzy sets that represent the outputs of each rule are combined into a single fuzzy set. Aggregation only occurs once for each output variable, just prior to the fifth and final step, defuzzification. The input of the aggregation process is the list of truncated output functions returned by the implication process for each rule. The output of the aggregation process is one fuzzy set for each output variable.

5. In "Biological Neural Network" and " Computer" which is better in terms of the speed. Also explain with reason. (03 Marks)

Biological Neural Networks = Computers

Speed = Fast (nanoseconds) = Slow (milliseconds)

Processing = Superior (massively parallel) = Inferior (Sequential mode)

Size & Complexity = 10¹¹ neurons, 10¹⁵ interconnections = Far few processing elements

Storage = Adaptable, interconnection strengths = Strictly replaceable

Fault tolerance = Extremely Fault tolerant = Inherently non fault tolerant

Control mechanism = Distributive control = Central control

Reason = While this clearly shows that the human information processing system is superior to conventional computers, but still it is possible to realize an artificial neural network which exhibits the above mentioned properties.

What is Un-supervised learning. Explain karan tha. (03 Marks)

Unsupervised: ♣ Given a set of examples with no labeling, group them into sets called clusters

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Supposed you are in the Knowledge Analysis Phase, and the Domain Expert is required. What are the difficulties faced by you with the Domain Expert in Knowledge analysis phase. (05 Marks)

In the Knowledge Analysis Phase, collaborating with a Domain Expert is crucial for understanding the specific domain, its intricacies, and the requirements of the project. However, several difficulties can arise during this phase:

Communication Gap: Time Constraints: Differing Perspectives: Domain Complexity: Limited Availability: Resistance to Change: Domain Expert Bias:

The different programming languages have some data types defined. What are the 5 defined data types of CLIPS language. (05 Marks)

The CLIPS programming language defines eight primitive data types for representing information. These types are

Float
Integer
Symbol
String
External Address
Fact Address
Instance Name
Instance Address

Any three advantages of Artificial Neural Networks. (3)

Advantages

- Excellent for pattern recognition
- Excellent classifiers
- Handles noisy data well
- Good for generalization

Elaborate the concept of short term and long term memory in Expert Systems. (3)

STM is used for temporary storage and manipulation of information during reasoning, while LTM is used for long-term storage of the knowledge base. The interaction between STM and LTM is essential for the functioning of expert systems, as it allows them to process information dynamically while maintaining a comprehensive knowledge base for future use.

Names of three components of Action predicates. (3)

Action is a predicate used to change states. It has three components namely, the predicate itself, the pre-condition, and post-condition predicates. For example, the action to buy something item can be represented as,

Action:

buy(X)

Pre-conditions:

at(Place) $\wedge$ sells(Place, X)

Post-conditions/Effect:

have(X)

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A student wants to understand the concept of Expert Systems. Elaborate properly the domains of Expert Systems. (5)

Expert systems are computer programs that use artificial intelligence to simulate the judgment and behavior of human experts. They are designed for specific domains such as medicine, science, finance, and more, where experts are required to make diagnoses, judgments, or predictions

Any five applications of Fuzzy System. (5)

Five parts of the fuzzy inference process

- Fuzzification of the input variables
- Application of fuzzy operator in the antecedent (premises)
- Implication from antecedent to consequent
- Aggregation of consequents across the rules
- Defuzzification of output

Applications:

- 1) Control Systems: Used in air conditioners, washing machines, and car systems for smooth, efficient operation.
- 2) Pattern Recognition: Helps in image, speech, and handwriting recognition by handling unclear boundaries between patterns.
- 3) Automotive Industry: Used in automatic transmissions, engine controls, and ABS for better performance and safety.
- 4) Consumer Electronics: Applied in cameras, washing machines, and rice cookers for improved functionality and user experience.
- 5) Medical Diagnosis: Assists in diagnosing diseases and recommending treatments by considering the uncertainty in symptoms and test results.

Any five applications where Learning Algorithms are used. (5)

- Spoken digits and word recognition
- Handwriting recognition
- Driving autonomous vehicles
- Path finders
- Intelligent homes
- Intrusion detectors
- Intelligent refrigerators, tvs, vacuum cleaners
- Computer games
- Humanoid robotics

write down the command add following facts

Animal(sheep)

owner(Ali)

Solution

(assert (animal sheep))

(assert (owner Ali))

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Linearly non-separable and linearly separable explain with example

Linearly separable data refers to a set of input vectors for which a hyperplane can be found such that each vector lies on the pre-assigned side of the hyperplane. This means that the data can be classified using a linear decision boundary. For example, consider the logical AND operation, where input vectors (1, 1) and (0, 0) belong to one class, and (1, 0) and (0, 1) belong to another class. A hyperplane can be drawn to separate these points into two classes, making the data linearly separable.

On the other hand, linearly non-separable data cannot be classified using a linear decision boundary. For instance, consider the XOR problem, where input vectors (0, 0) and (1, 1) belong to one class, and (0, 1) and (1, 0) belong to another class. No hyperplane can be drawn to separate these points into two classes, making the data linearly non-separable.

Increase in number of combination in an instance space has no effect on number of combination in a concept space. Do you agree with the statement or not given reasons to support your answer. Shortly

I don't agree with the statement. The number of combinations in an instance space and concept space are related, especially in the context of machine learning and pattern recognition.

Reason, an increase in the number of combinations in an instance space can lead to a corresponding increase in the complexity of the concept space as the learning algorithm tries to capture and generalize from the increased variability in the data.

Which type of ordering in hypothesis space is best suited for a situation we start from $h < ; >$ hypothesis and move toward $s < 0, 0 >$ hypothesis

In the context where you start from the hypothesis $h < ; >$ and move toward the hypothesis $s < 0, 0 >$, it seems you're moving from a more general hypothesis towards a more specific one. This suggests a specialization or refinement ordering in the hypothesis space. In general, this type of ordering is suited for situations where you start with a broad hypothesis and iteratively refine it to fit the data or problem constraints more closely.

How can you elaborate the prime role of an expert system designer

One of the prime roles of the expert system designer is to act as a knowledge engineer. As a knowledge engineer, the designer must overcome the knowledge acquisition bottleneck and find an effective way to get information from the expert and encode it in the knowledge base, using one of the knowledge representation techniques we discussed in KRR.

Briefly explain the defects construct in CLIPS with example

The DEFFACTS construct

These are a set of facts that are automatically asserted when the (reset) command is used, to set the working memory to its initial state. For example:

```
CLIPS> (def facts myFacts "My known facts
( man ( name "Ahmed" ) )
(father
(fathersName "Ahmed")
(sonsName "Belal" ) ) )
```

Write 3 knowledge acquisition techniques

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- Knowledge elicitation by interview
- Electronic brainstorming
- On-site observation

Write clips program to print “hello world”

```
clips
(deffacts hello-world ()
  (fact))
(defrule print-hello-world
  (fact)
=>
  (printout t "Hello World!\n")
  (retract fact))
(reset)
(run)
```

Define a template using deftemplate to add the following fact to it: CLIPS>(assert (father (fathersName “Ahmed”) (sonsName “Belal”)))

Solution

```
(deftemplate father
  (slot fathersName (type SYMBOL))
  (slot sonsName (type SYMBOL))
)
```

```
(assert (father (fathersName "Ahmed") (sonsName "Belal")))
```

Which inference mechanism is well suited for design of automated diagnosis expert system.

Justify with reasons

The inference mechanism that is well suited for the design of an automated diagnosis expert system is backward chaining. Backward chaining is a type of deductive reasoning where the inference engine starts with the goal and works backward, looking for evidence that supports that goal. This method is particularly useful when the possible conclusions are limited in number and well defined, such as in classification or diagnosis type systems, in which each of several possible conclusions can be checked to see if it is supported by the data.

Which term is used to describe the problems for which data cannot be separated into their respective classes by using a single straight line. Also give an example

The term used to describe problems where data cannot be separated into their respective classes by using a single straight line is "non-linear separability." In such cases, the classes are not linearly separable, meaning they cannot be separated by a single straight line in the feature space. One example of a problem with non-linear separability is the XOR (exclusive OR) problem. In a 2D space, the XOR problem consists of four points: (0,0) and (1,1) belonging to one class, and (0,1) and (1,0) belonging to the other class. It is not possible to draw a single straight line that separates these two classes.

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In the context of machine learning, do you agree with “If it makes a mistake once it will always make the same mistake every time it runs”. Justify with reasons

No, I don't agree with the statement "If it makes a mistake once, it will always make the same mistake every time it runs" in the context of machine learning.

Reason, while machine learning models strive to be consistent and generalize well, they are not deterministic in the sense that they will always make the same mistake every time they run.

Variability in the data, model, and algorithmic factors can lead to different mistakes being made in different runs.

Convert to CNF form: $(A \rightarrow B) \rightarrow B$

Given expression: $(A \rightarrow B) \rightarrow B$

Step 1: Eliminate the implication

$(A \rightarrow B) \rightarrow B$ becomes $(\neg A \vee B) \rightarrow B$

Step 2: Apply the equivalence $X \rightarrow Y \equiv \neg X \vee Y$ to the outer implication

$(\neg A \vee B) \rightarrow B$ becomes $\neg(\neg A \vee B) \vee B$

Step 3: Apply De Morgan's law $\neg(X \vee Y) \equiv \neg X \wedge \neg Y$ to negate the inner disjunction

$\neg(\neg A \vee B) \vee B$ becomes $(A \wedge \neg B) \vee B$

Step 4: Apply the distributive property of $\vee$ over $\wedge$

$(A \wedge \neg B) \vee B$ becomes $(A \vee B) \wedge (\neg B \vee B)$

Step 5: Simplify $\neg B \vee B$ to 1 (true)

$(A \vee B) \wedge (\neg B \vee B)$ becomes $(A \vee B) \wedge 1$

Step 6: Simplify $A \vee B$ to 1 (true)

$(A \vee B) \wedge 1$ becomes 1

So, the CNF form of $(A \rightarrow B) \rightarrow B$ is 1, which is a tautology (always true).

Applications of soft computing.

Handwritten Script Recognition.

Image Processing and Data Compression.

Automotive Systems and Manufacturing.

Soft computing based Architecture.

Decision Support System.

Power System Analysis.

Bioinformatics.

Investment and Trading.

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3 drawbacks of elicitation .

Bias: Elicitation can be biased by the knowledge, beliefs, or preferences of the person conducting the elicitation or the experts being elicited, leading to inaccurate or skewed results.

Incomplete Information: Elicitation relies on the expertise and knowledge of individuals, which may be limited or incomplete, leading to gaps in the understanding of the problem domain or solution space.

Subjectivity: Elicitation is often subjective, as it involves interpreting and synthesizing information from different sources, which can lead to varying interpretations and conclusions. This subjectivity can introduce uncertainty and make it challenging to reach consensus.

Crisp or fuzzy ka difference with example

crisp sets have clear boundaries, while fuzzy sets have blurred boundaries. Crisp sets are commonly used in traditional computing, while fuzzy sets are used in soft computing.

Connectionist ki applications.

Although ID3 spanned more of the concept space, but still there is a possibility that the true concept is not simply a mixture of disjunctions of conjunctions, but some more complex arrangement of attributes.

Expert system explain krna tha.

Expert Systems (ES) are a popular and useful application area in AI. Having studied KRR, it is instructive to study ES to see a practical manifestation of the principles learnt there

Knowledge engineer

A knowledge engineer is 'a person who designs, builds and tests an Expert System' (Durkin). A knowledge engineer plays a key role in identifying, acquiring and encoding knowledge.

Degree of truth

Membership Function (μ)

The degree of truth that we have been talking about, is specifically driven out by a function called the membership function. It can be any function ranging from a simple linear straight line to a complicated spline function or a polynomial of a higher degree.

Clustering

Clustering is a form of unsupervised learning, in which the training data is available but without the classification information or class labels. The task of clustering is to identify and group similar individual data elements based on some measure of similarity.

Inference mechanism of forward and backward chaining

Forward Chaining

It is an "inference strategy that begins with a set of known facts, derives new facts using rules whose premises match the known facts, and continues this process until a goal state is reached or until no further rules have premises that match the known or derived facts" (Durkin).

Backward chaining

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Backward chaining is an inference strategy that works backward from a hypothesis to a proof. You begin with a hypothesis about what the situation might be.

Three phase in machine learning

1. Training: a training set of examples of correct behavior is analyzed and some representation of the newly learnt knowledge is stored. This is often some form of rules.
2. Validation: the rules are checked and, if necessary, additional training is given. Sometimes additional test data are used, but instead of using a human to validate the rules, some other automatic knowledge based component may be used. The role of tester is often called the critic.
3. Application: the rules are used in responding to some new situations.

Genetic algorithm wala ta

Genetic algorithms is a modern advancement to the hill climbing search based problem solving. Genetic algorithms are inspired by the biological theory of evolution and provide facilities of parallel search agents using collaborative hill climbing.

CLIPS ki print command

The PRINTOUT command

Instead of asserting facts in a rule, you can print out messages using (printout t "Ali is Ahmed's son" crlf)

A person wants to know about the Artificial Neural Networks. How can you elaborate the design phases of Artificial Neural Networks? (5 Marks)

Design phases of ANNs

- Feature Representation
- The number of features are determined using no of inputs for the problem.
- Training
- Training is either supervised or unsupervised
- Similarity Measurement
- A measure to tell the difference between the actual output of the network while training and the desired labeled output
- Validation
- During training, training data is divided into k data sets; k-1 sets are used for training, and the remaining data set is used for cross validation.
- Stopping Criteria
- Done through MSE. We define a low threshold usually 0.01, which if reached stops the training data.
- Application Testing
- A network is said to generalize well when the input-output relationship computed by the network is correct (or nearly so) for input-output pattern (test data) never used in creating and training the network.

Supervised

Given a set of example input/output pairs, find a rule that does a good job of predicting the output associated with a new input.

Fuzzy set

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Fuzzy sets, unlike classical sets, do not restrict themselves to something lying wholly in either set A or in set not-A. They let things sit on the fence, and are thus closer to the human world.

Use Clips command to show product of two numbers ... number was given

```
(defrule calculate-product
```

```
(numbers ?x ?y)
```

```
=>
```

```
(bind ?product (* ?x ?y))
```

```
(printout t "The product of " ?x " and " ?y " is " ?product crlf)
```

```
)
```

```
(assert (numbers 5 10))
```

Forward Chaining disadvantages

Eliminate and synchronizing available data may consume a lot of time, which may stretch the process. The explanation of facts or observations in forward chaining may not be very clear.

Clips command used to add and modify facts

We add a fact:

```
CLIPS>(assert ( father ( fathersName "Ahmed") (sonsName "Belal") ) )
```

To modify the fathers name slot, enter the following:

```
CLIPS> (modify 2 ( fathersName "Ali Ahmed"))
```

Linear model phases

- Planning
- Knowledge acquisition and analysis
- Knowledge design
- Code
- Knowledge verification
- System evaluation

Fuzzy logic partially true h statement s agree krt ya nhi btana tha reason k sath

Fuzzy logic is often described as "partially true" because it allows for the representation of degrees of truth. In classical (Boolean) logic, statements are either true or false, with no in-between. However, in many real-world situations, the truth of a statement is not black or white but exists on a spectrum of possibilities. Fuzzy logic provides a way to handle this kind of uncertainty by allowing statements to have a degree of truth between 0 and 1.

Information gain k que tha

Given entropy as a measure of the impurity in a collection of training examples, we can now define a measure of the effectiveness of an attribute in classifying the training data. The measure we will use, called information gain.

Entropy

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In order to define information gain precisely, we begin by defining a measure commonly used in statistics and information theory, called entropy

Define Computer Vision.

The purpose of computer vision is to study algorithms, techniques and applications that help us make machines that can "understand" images and videos.

Three stages of learning.

Rote learning

Deductive learning

Inductive learning

Formulas of AND,OR,NOT gate in fuzzy logics.

1) AND = $\min(A,B)$ 2) OR $\max(A,B)$ 3) NOT $1 - A$

Write Steps of linear method while developing Expert System.

Problem Identification: Define the problem the expert system will solve.

Knowledge Acquisition: Gather expertise from human experts.

Knowledge Representation: Formalize acquired knowledge for computer understanding.

Knowledge Base Development: Create the knowledge base using the formalized knowledge.

Inference Engine Development: Develop the engine for reasoning and decision-making.

User Interface Development: Create an interface for user interaction.

Testing and Validation: Test the system for accuracy and effectiveness.

Deployment and Maintenance: Deploy the system and maintain it over time.

Evaluation: Assess the system's performance and impact.

Do you agree *planning predicate* define state and *Condition predicate* change State? Give reason.

Planning predicates define the state of the world, while condition predicates define the conditions that must be met for an action to be executed, which may change the state of the world.

Difference Between Boolean and Fuzzy

Boolean logic is a two-valued logic that deals with reasoning that is fixed and exact, while fuzzy logic is a multi-valued logic that deals with reasoning that is approximate rather than fixed and exact. Boolean logic is used in traditional computing, while fuzzy logic is used in soft computing and is more suitable for modeling real-world situations where things are not black and white

Fuzzy vs. probability

Fuzzy logic and probability theory are complementary approaches to uncertainty, and they are not interchangeable. Fuzzy logic is a subset of fuzzy set theory, while probability theory is a subset of classical set theory.

Crisp is a Boolean two-valued function, is the statement correct? Support your answer with example. (3 Marks)

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Yes, the statement that "Crisp is a Boolean two-valued function" is correct. A crisp function, also known as a Boolean function or a binary function, is a function that can have only two possible values, typically represented as 0 (false) and 1 (true) in the context of Boolean algebra

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